PlanktonScope Statement of Need

Purchase Request 143041-25-0034

I. Purpose: To purchase a moored shadowgraph camera system that will provide data on the prey important for protected species. Data from this camera system will augment existing ship-based sampling with higher temporal resolution data. This will be the first plankton camera system moored on the West coast in the California Current System, and the engineering and deployment of this prototype will help colleagues at the AFSC develop a similar system. The camera will provide real-time imagery and taxonomically resolved abundance estimates of plankton prey that provide information on ocean conditions and prey for protected species.

II. Technical Specifications:

PlanktonScope|Mooring profiler

Pressure rating: 250m

- High-pressure resistant chamber (2)
- Sapphire glass window (2)
- Illumination Unit (1)
- High-frequency grayscale imaging camera (1)
- Embedded computer system
- Solid state hard drive (2TB)
- Marine radial wiper system (1)
- Subconnectors (8)
- Feature: Integrated UI
- Accessories: L-Key set, Super-Corrosion-Resistant 316 Stainless Steel Socket Head Screws (2*6), Molykote medium lubricant 44 (1), O-rings (2*4)

AI-based Plankton Recognition and Enumeration System

- Graphic Processing Unit (1)
- PlanktonScope Identification Software|Ver 3.2
- Pretrained classification models

Real-time monitoring and visualization software

- Platform for monitoring instrument status
- Dynamic database for store processed data
- Real time platform for live stream and processed data display

III. Deliverables:

- PlanktonScope
- AI-based Plankton Recognition and Enumeration System
- Real-time monitoring and visualization software

IV. Requested Deliverable Schedule: October 31, 2025

V. Place of Delivery:

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